

Week 7 Cost-Benefit Memo: Is a More Complex Model Worth It?

To: Dr. De Fretias, ED Board, Martina Griffith (Clinical IT Lead) **From:** [Your Name] **Re:** Random Forest vs. Logistic Regression Baseline — Recommendation for Phase 3

Azaria DaSilva

Verdict

Do not adopt the random forest. The Week 6 logistic regression baseline remains the recommended model for Phase 3, it is more accurate, faster at inference, and considerably easier to explain to a clinician, and the added complexity of a random forest bought no improvement whatsoever on the one metric that matters most: catching the sickest patients.

Dataset and Methods Recap

This benchmark reuses the Week 5–6 Yale EMLLC triage extract (55,121 adult ED encounters) and the exact same 80/20 stratified train/test split (`random_state=42`) established in Week 6, so results are directly comparable. Four models were evaluated: a stratified dummy baseline, the Week 6 logistic regression and decision tree baselines, and a new Random Forest classifier (200 trees, max depth 10) as this week's complex-model candidate. All models were trained and evaluated on the same feature set — vital signs and chief-complaint flags, with demographic fields and outcome-leakage columns (`disposition`, `previousdispo`) excluded throughout.

Benchmark Table

Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)	Train Time (s)	Inference (ms/patient)	Recall (ESI 1)
Dummy Baseline	0.370	0.201	0.201	0.201	—	—	0.000
Logistic Regression	0.683	0.614	0.498	0.533	11.178	0.0035	0.400
Decision Tree	0.553	0.242	0.249	0.224	0.259	0.0029	0.000
Random Forest	0.587	0.333	0.267	0.245	4.962	0.101	0.000

Logistic regression leads on every accuracy-based metric and, critically, is the only model to catch any ESI Level 1 patients at all (recall 0.400, versus 0.000 for both the decision tree and the random forest).

Interpretability

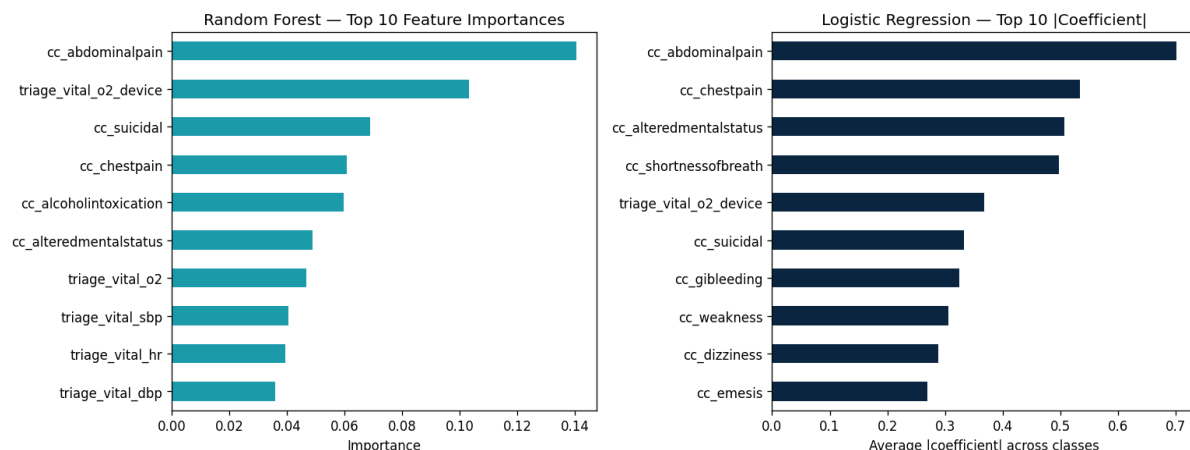


Figure 1: Random Forest feature importances (left) vs. Logistic Regression coefficient magnitudes (right). Both models agree that *cc_abdominalpain*, *cc_chestpain*, and *triage_vital_o2_device* are strong signals.

Logistic regression: explainable in under a minute. Each feature carries one fixed, signed coefficient per class, so a single prediction can be explained in one sentence — "this patient was flagged as higher acuity mainly due to abdominal pain and altered mental status." The explanation method is identical for every patient.

Random forest: not explainable in under a minute. Feature importances describe average behaviour across all 55,121 patients, not the reasoning behind any one specific prediction, which is instead the averaged vote of 200 individual trees. Explaining a single prediction properly would require a tool such as SHAP — a legitimate further analysis step, but not a bedside conversation.

Three Arguments for the Recommended Choice (Logistic Regression)

1. **It is the only model that catches any critical patients.** A 0.400 recall on ESI Level 1 is still far from adequate, but it is meaningfully better than the 0.000 achieved by both the decision tree and the random forest — the two models notionally more sophisticated than it.
2. **It is roughly 29 times faster at inference** than the random forest (0.0035 ms vs. 0.101 ms per patient). In a real-time triage setting, this difference compounds across every patient, every shift, indefinitely.
3. **It is transparent by construction.** Every prediction can be explained to a clinician in one sentence, with the same reasoning applied consistently to every patient — a requirement Dr. Reyes has stated explicitly, not a nice-to-have.

Three Arguments Against the Recommended Choice

1. **Its overall recall (macro) is still modest (0.498)** — half of all triage-level predictions across all classes are, on average, missed. It is the best baseline available, not a solution.
2. **It required the longest training time in this benchmark (11.2 seconds)**, though this is a one-off cost paid during model development, not during deployment, so its practical weight is low.
3. **It is a linear model**, and cannot capture interactions between features that a tree-based method might, in principle, exploit — though this benchmark found no evidence that the random forest actually exploited any such interaction to clinical benefit.

Risks and Unknowns

- **We do not yet know whether the random forest's failure on ESI Level 1 is a fundamental data limitation (too few ESI 1 cases, 77 in the full dataset) or a fixable hyper-parameter issue (e.g. class weighting, different tree depth).** This benchmark does not rule out the possibility that a properly tuned random forest could still outperform logistic regression on this metric.
- **Neither model is yet fit for clinical deployment.** A 0.400 recall on the most critical patient class means 6 in 10 ESI Level 1 patients would still be missed under the current logistic regression baseline. This memo recommends logistic regression as the *better of the two options benchmarked*, not as a deployment-ready system.

Recommendation

Proceed to Phase 3 with **logistic regression**, not the random forest. Future development effort should focus on techniques that directly address class imbalance for the existing linear model, such as class weighting or a cost-sensitive decision threshold tuned specifically to raise ESI Level 1 recall — rather than pursuing additional model complexity that has not, in this benchmark, delivered any clinical benefit.